

| Name                             | Description                              | Rationale                                                                                                                                                 | Priority (1[highest]-3[lowest]) | Author                                                                                                                                   |                |  |
|----------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------|--|
| FUNCTIONAL REQUIREMENTS          |                                          |                                                                                                                                                           |                                 |                                                                                                                                          |                |  |
| Account Management               |                                          |                                                                                                                                                           |                                 |                                                                                                                                          |                |  |
| 1                                | Account Creation                         | The user shall create an account.                                                                                                                         | 1                               | An account is necessary to access the features of the program.                                                                           | Robert Albetel |  |
| 2                                | Guest Access                             | The user shall submit an urgent relief request without creating an account.                                                                               | 1                               | Residents in an emergency may not have time to complete registration.                                                                    | Tavien Smith   |  |
| 3                                | Account Roles                            | The system shall classify each account as one of the following roles: Resident, Volunteer, Shelter Operator, or Coordinator.                              | 1                               | Distinct roles let the platform show each user the tools relevant to their part in the relief effort.                                    | Tavien Smith   |  |
| 4                                | Default Account Type                     | The system shall classify accounts as resident accounts by default.                                                                                       | 1                               | Defaulting to a resident account still allows the user to register as a volunteer later on their own time.                               | Robert Albetel |  |
| 5                                | Account Name                             | The system shall record the account name.                                                                                                                 | 1                               | The resident's name is useful for identifying them later.                                                                                | Robert Albetel |  |
| 6                                | Account Address                          | The system shall record the account address every 30 minutes.                                                                                             | 2                               | The address is important for navigating to the emergency.                                                                                | Robert Albetel |  |
| 7                                | User Verification                        | The user shall verify their identity to become a verified user through a background check, one-time passcode, or linked third-party account.              | 2                               | Verification prevents users from requesting emergency help as a prank, wasting responders' time.                                         | Zaki Bushiri   |  |
| Needs Reporting & Tracking       |                                          |                                                                                                                                                           |                                 |                                                                                                                                          |                |  |
| 8                                | Relief Request Submission                | The user shall make a request for relief.                                                                                                                 | 1                               | Relief requests are the cornerstone of the program and how volunteers track emergencies.                                                 | Robert Albetel |  |
| 9                                | Request Status Updates                   | The user shall update the status of their request as needed.                                                                                              | 2                               | The status of a request changes as it is worked on, so updates allow for adjustments.                                                    | Robert Albetel |  |
| 10                               | Request Status Visibility                | The system shall provide a means for users to view the status of a submitted request.                                                                     | 1                               | Victims and volunteers need to know a request's status so multiple volunteers don't respond to one, and so victims know help is coming.  | Zaki Bushiri   |  |
| 11                               | Duplicate Request Handling               | The system shall combine duplicate requests from the same location.                                                                                       | 1                               | Users in a panicked state may resend a request they think didn't go through, wasting time and resources if a unit is already dispatched. | Zaki Bushiri   |  |
| 12                               | Emergency Priority Classification        | The system shall determine the priority level of a request based on the contents of the report.                                                           | 1                               | This lets the system help those in life-threatening situations faster than someone who is less urgently affected.                        | Zaki Bushiri   |  |
| 13                               | Number Needing Help                      | The user shall be prompted to enter the number of people needing help.                                                                                    | 1                               | This prevents sending too many or too few volunteers to a location.                                                                      | Zaki Bushiri   |  |
| 14                               | Request Cancellation                     | The user shall cancel their request.                                                                                                                      | 2                               | If the user is safe or already helped by someone else, cancelling lets the team move on to others in need.                               | Zaki Bushiri   |  |
| 15                               | Request Detail Display                   | The system shall display additional details about a request.                                                                                              | 2                               | Further information provided after the initial request is useful for coordinating and adapting to the situation.                         | Robert Albetel |  |
| 16                               | Flagging Suspicious Requests             | The system shall flag a request as suspicious for coordinator review.                                                                                     | 2                               | This prevents resources from being spread thin on bad-faith or invalid requests.                                                         | Zaki Bushiri   |  |
| Volunteer Coordination           |                                          |                                                                                                                                                           |                                 |                                                                                                                                          |                |  |
| 17                               | Volunteer Specialties & Equipment        | The user shall be given a space to list their specialties or available equipment.                                                                         | 2                               | Specialties such as medical training or equipment such as saws will be vital for certain tasks.                                          | Robert Albetel |  |
| 18                               | Volunteer Availability Status            | The user shall set their volunteer availability status.                                                                                                   | 1                               | This allows for a more accurate count of volunteers on hand at any given moment.                                                         | Zaki Bushiri   |  |
| 19                               | Volunteer Count Tracking                 | The system shall track the number of available volunteers every 5 minutes.                                                                                | 2                               | Knowing volunteer counts helps decide who to send where and how many people can be spared.                                               | Robert Albetel |  |
| 20                               | Volunteer Request List                   | The user shall view a list of relief requests when acting as a volunteer.                                                                                 | 1                               | Volunteers need to know when something happens and where, along with other details.                                                      | Robert Albetel |  |
| 21                               | Volunteer Self-Assignment                | The user shall accept a request directly as a volunteer instead of waiting on a coordinator.                                                              | 1                               | This increases the number of people who can be helped, since a nearby volunteer can respond even out of turn.                            | Zaki Bushiri   |  |
| 22                               | Volunteer Request Claiming               | The user shall claim the requests they choose to respond to when acting as a volunteer.                                                                   | 1                               | Claiming requests helps track which situations are being handled and how many others still need help.                                    | Robert Albetel |  |
| Shelter Management               |                                          |                                                                                                                                                           |                                 |                                                                                                                                          |                |  |
| 23                               | Nearby Shelter List                      | The system shall include a list of all shelters near the account's location.                                                                              | 1                               | Finding a shelter is one of the most important features of the program.                                                                  | Robert Albetel |  |
| 24                               | Shelter Proximity Search                 | The user shall search shelters by proximity to their current location.                                                                                    | 1                               | Residents need to find the closest available shelter quickly during an emergency.                                                        | Tavien Smith   |  |
| 25                               | Shelter Search Radius                    | The user shall specify their search radius when selecting a shelter.                                                                                      | 1                               | Users looking for closer or more suitable shelters should be able to specify a range.                                                    | Robert Albetel |  |
| 26                               | Shelter Filtering by Accommodation       | The user shall filter shelter search results by accommodation criteria such as pet-friendly, wheelchair-accessible, or medical-support-available.         | 2                               | Not all shelters accommodate every need; filtering prevents users from arriving at a shelter that can't serve them.                      | Nate Sheffield |  |
| 27                               | Shelter Data Freshness                   | The system shall show when a shelter's inventory, capacity, or route information was last updated.                                                        | 1                               | Information changes quickly during hurricanes, so users should know whether a listing is current.                                        | Zaki Bushiri   |  |
| 28                               | Shelter Info Editing                     | The system shall allow changes to a shelter's information by an authorized user.                                                                          | 2                               | If a shelter restocks a resource or changes a policy such as pet allowance, that information must stay current.                          | Robert Albetel |  |
| 29                               | Crowdsourced Shelter Status Verification | The user shall flag a shelter as full or closed, which the system shall verify against official shelter data before removing it from the active list.     | 1                               | Crowdsourced reports speed up detection of capacity changes, but unverified reports could mislead other users.                           | Nate Sheffield |  |
| 30                               | Live Evacuation Routes                   | The system shall display real-time evacuation routes overlaid on a map, updated at least every 15 minutes based on road closure data.                     | 1                               | Evacuation routes can become blocked by flooding or debris; static routes risk directing users into unsafe conditions.                   | Nate Sheffield |  |
| Shelter Partnerships & Donations |                                          |                                                                                                                                                           |                                 |                                                                                                                                          |                |  |
| 31                               | Shelter Partnership Application          | The user shall submit a partnership application to register a shelter with the network.                                                                   | 2                               | Shelters need a formal way to join the network rather than being added informally.                                                       | Tavien Smith   |  |
| 32                               | Partnership Application Review           | The user shall review a pending shelter partnership application when acting as a coordinator.                                                             | 2                               | Coordinators need to vet a shelter before it appears to residents.                                                                       | Tavien Smith   |  |
| 33                               | Relief Organization Supply Posts         | The system shall allow a verified relief organization account to post a supply-need update that appears in the donation feed within 5 minutes of posting. | 2                               | Real-time supply-need visibility reduces wasted donations of items shelters don't actually need.                                         | Nate Sheffield |  |

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| 34                                          | Donation & Volunteer Opportunity Listings | The user shall view a list of nearby donation drop-off points or volunteer sign-up opportunities, filterable by distance or need type.                                                          | 2 | Relief efforts depend on efficient matching of willing donors and volunteers with actual on-the-ground needs.                        | Nate Sheffield   |  |
| <b>Communication &amp; Notifications</b>    |                                           |                                                                                                                                                                                                 |   |                                                                                                                                      |                  |  |
| 35                                          | Resident-Volunteer Communication          | The system shall provide a direct communication channel linking a resident with the volunteer assigned to their request.                                                                        | 2 | A direct channel lets volunteers and residents share real-time updates about the situation.                                          | Robert Albetel † |  |
| 36                                          | Impact Zone Push Notifications            | The system shall send a push notification to all users within a projected impact zone when the National Weather Service issues a new advisory for that zone, within 90 seconds of issuance.     | 1 | Timely alerts can be the difference between safe evacuation and being caught in the storm.                                           | Nate Sheffield   |  |
| 37                                          | Selective Notification Muting             | The user shall mute non-critical notifications while critical alerts remain non-mutable.                                                                                                        | 2 | Notification fatigue can cause users to disable alerts entirely; selective muting keeps life-safety alerts guaranteed to reach them. | Nate Sheffield   |  |
| <b>Emergency &amp; Safety Tools</b>         |                                           |                                                                                                                                                                                                 |   |                                                                                                                                      |                  |  |
| 38                                          | SOS Button Activation                     | The user shall activate an SOS button that transmits the user's GPS coordinates to local emergency services within 10 seconds of activation, provided network connectivity exists.              | 1 | In life-threatening situations, speed of contact with responders is critical.                                                        | Nate Sheffield   |  |
| 39                                          | Deferred SOS Transmission                 | The system shall queue an SOS request locally when no connectivity is available, then transmit it automatically within 30 seconds of connectivity being restored.                               | 1 | Loss of signal shouldn't mean loss of the emergency request.                                                                         | Nate Sheffield   |  |
| 40                                          | One-Tap Safe Status                       | The system shall allow a verified user to mark themselves as safe with one tap, immediately notifying their designated emergency contacts of their status, including their last known location. | 3 | Phone networks are often congested during a storm; this lets a user signal safety without straining the network.                     | Zaki Bushiri     |  |
| 41                                          | Emergency Contact Storage                 | The user shall save emergency contact information that can be accessed quickly during a hurricane.                                                                                              | 3 | Users may need to quickly reach family, emergency services, or shelters without searching for their information.                     | Zaki Bushiri     |  |
| 42                                          | Emergency Contact Notifications           | The user shall add up to 10 emergency contacts, each of whom shall receive an automated notification when the user updates their safety status.                                                 | 2 | Family members outside the affected area need a reliable way to learn a loved one's status.                                          | Nate Sheffield   |  |
| 43                                          | Hazard Reporting                          | The user shall report a hazard by providing a photo with a GPS location, which the system shall display to other users within 5 minutes of submission.                                          | 2 | Community-sourced hazard reports fill gaps between official updates, which can lag behind fast-changing conditions on the ground.    | Nate Sheffield † |  |
| 44                                          | Address Privacy                           | The system shall hide a resident's exact address from a volunteer until the volunteer accepts their request.                                                                                    | 1 | Withholding exact addresses until a match protects resident privacy from unverified users.                                           | Tavien Smith     |  |
| 45                                          | Location Tracking Opt-Out                 | The user shall opt out of location tracking while retaining access to manually searchable shelter or alert information.                                                                         | 2 | Some users may have privacy concerns; core safety features shouldn't require surrendering location data.                             | Nate Sheffield   |  |
| <b>Storm &amp; Preparedness Information</b> |                                           |                                                                                                                                                                                                 |   |                                                                                                                                      |                  |  |
| 46                                          | Severe Weather Alerts                     | The user shall receive alerts when a hurricane or severe weather event is approaching their area.                                                                                               | 1 | Alerts give users enough time to prepare, evacuate, or seek shelter before conditions become dangerous.                              | Zaki Bushiri     |  |
| 47                                          | Storm Data Integration                    | The system shall integrate with the National Weather Service API to display current storm conditions, refreshed at least every 10 minutes.                                                      | 1 | Authoritative, frequently updated storm position, path, and wind-speed data is essential for time-sensitive evacuation decisions.    | Nate Sheffield † |  |
| 48                                          | Emergency Instructions                    | The system shall display instructions specific to the type of emergency being reported.                                                                                                         | 2 | This educates users on how to respond to different emergencies, helping them stay out of dangerous situations.                       | Zaki Bushiri     |  |
| 49                                          | Preventative Measure Guidance             | The system shall provide instructions for preventative measures to take before a hurricane arrives.                                                                                             | 2 | The platform is meant to serve both as a relief center and a resource for reducing hurricane damage.                                 | Robert Albetel   |  |
| <b>Offline Access &amp; Resilience</b>      |                                           |                                                                                                                                                                                                 |   |                                                                                                                                      |                  |  |
| 50                                          | Offline Data Access                       | The system shall cache recently downloaded shelter data, evacuation routes, plus emergency contact numbers locally for offline access.                                                          | 1 | Cellular and Wi-Fi networks often fail during hurricanes; offline access ensures critical information remains usable.                | Nate Sheffield † |  |
| 51                                          | Offline Communication                     | The system shall allow structured communication between users to continue functioning without an internet connection.                                                                           | 1 | During disasters, internet connection may be unavailable, so it is important to plan for this contingency.                           | Zaki Bushiri     |  |
| 52                                          | Offline Map Downloads                     | The user shall download map regions offline containing emergency locations.                                                                                                                     | 1 | Power and cell towers often go out or are damaged during hurricanes.                                                                 | Zaki Bushiri     |  |
| 53                                          | Offline Location Snapshot                 | The system shall snapshot the user's latest known location in case internet access is lost.                                                                                                     | 1 | During disasters, internet connection may be unavailable, so it is important to plan for this contingency.                           | Zaki Bushiri     |  |
| 54                                          | Offline First-Aid Instructions            | The system shall display basic first-aid instructions accessible without an internet connection.                                                                                                | 2 | Medical help may be delayed for hours; offline first-aid guidance can prevent minor injuries from worsening.                         | Nate Sheffield   |  |
| <b>Coordinator Tools</b>                    |                                           |                                                                                                                                                                                                 |   |                                                                                                                                      |                  |  |
| 55                                          | Coordinator Dashboard                     | The user shall view all open requests sorted by urgency level when acting as a coordinator.                                                                                                     | 1 | Coordinators need to prioritize the most urgent unresolved needs first.                                                              | Tavien Smith     |  |
| 56                                          | Coordinator Request Search                | The user shall search open requests by need type when acting as a coordinator.                                                                                                                  | 2 | Searching lets a coordinator quickly find requests relevant to a specific type of resource they have available.                      | Tavien Smith     |  |
| 57                                          | Editing Resource Postings                 | The user shall edit or remove a resource posting submitted by an organization when acting as a coordinator.                                                                                     | 2 | Coordinators need the ability to correct inaccurate or outdated postings.                                                            | Tavien Smith     |  |
| <b>NON-FUNCTIONAL REQUIREMENTS</b>          |                                           |                                                                                                                                                                                                 |   |                                                                                                                                      |                  |  |
| <b>Usability Requirements</b>               |                                           |                                                                                                                                                                                                 |   |                                                                                                                                      |                  |  |
| 58                                          | Accessibility Compliance                  | The system shall meet WCAG 2.1 AA accessibility standards.                                                                                                                                      | 2 | Accessibility ensures residents with disabilities can use the platform during an emergency.                                          | Tavien Smith     |  |
| 59                                          | Browser Compatibility                     | The system shall be accessible through the most recent two versions of Chrome, Safari, or Firefox.                                                                                              | 3 | Supporting multiple browsers ensures access regardless of a user's device setup.                                                     | Tavien Smith     |  |
| 60                                          | Mobile App Availability                   | The system shall be available as a mobile application.                                                                                                                                          | 1 | Most users will access the platform from a phone during an evacuation or power outage.                                               | Tavien Smith     |  |
| 61                                          | Web Platform Availability                 | The system shall be available as a website.                                                                                                                                                     | 1 | The platform should be reachable without requiring an app download.                                                                  | Tavien Smith     |  |
| 62                                          | Low-Power Mode                            | The system shall switch to a low-power display mode when device battery falls below 20%.                                                                                                        | 2 | Phone charging is often unavailable for extended periods; conserving battery extends access to critical features.                    | Nate Sheffield   |  |
| <b>Performance Requirements</b>             |                                           |                                                                                                                                                                                                 |   |                                                                                                                                      |                  |  |
| 63                                          | Response Time                             | The system shall load the request submission page within 3 seconds under normal network conditions.                                                                                             | 2 | Slow load times could delay a resident from submitting an urgent request.                                                            | Tavien Smith     |  |

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| 64                                              | System Uptime                    | The system shall maintain 99.9% uptime during an active hurricane warning period, as measured by server availability monitoring. | 1 | The app is most critical exactly when infrastructure is most stressed.                                                      | Nate Sheffield   |  |
| 65                                              | Concurrent User Support          | The system shall support at least 5,000 concurrent users without service degradation.                                            | 1 | A hurricane can affect a large population at once, and the platform must not fail under peak demand.                        | Tavien Smith †   |  |
| <b>Maintainability and Support Requirements</b> |                                  |                                                                                                                                  |   |                                                                                                                             |                  |  |
| 66                                              | Data Backup                      | The system shall back up its database at least once every 24 hours.                                                              | 2 | Regular backups prevent permanent data loss in the event of a system failure.                                               | Tavien Smith     |  |
| <b>Security Requirements</b>                    |                                  |                                                                                                                                  |   |                                                                                                                             |                  |  |
| 67                                              | Data Encryption in Transit       | The system shall encrypt all sensitive user data while it is transmitted over the network.                                       | 1 | Location and medical-need data are sensitive; a breach could expose vulnerable users to risk beyond the disaster itself.    | Nate Sheffield   |  |
| 68                                              | Data Encryption at Rest          | The system shall encrypt all sensitive user data while it is stored.                                                             | 1 | Location and medical-need data are sensitive; a breach could expose vulnerable users to risk beyond the disaster itself.    | Nate Sheffield   |  |
| 69                                              | Account Lockout                  | The system shall lock a user account after five consecutive failed login attempts.                                               | 2 | Locking accounts after repeated failures protects against brute-force password attacks.                                     | Tavien Smith     |  |
| <b>Cultural Requirements</b>                    |                                  |                                                                                                                                  |   |                                                                                                                             |                  |  |
| 70                                              | Multi-Language Support (English) | The system shall support English for on-screen text with voice guidance.                                                         | 1 | English is the primary language of most current users.                                                                      | Nate Sheffield † |  |
| 71                                              | Multi-Language Support (Spanish) | The system shall support Spanish for on-screen text with voice guidance.                                                         | 1 | Hurricane-affected regions have significant Spanish-speaking populations who need equal access to safety information.       | Nate Sheffield † |  |
| 72                                              | Persistent Language Setting      | The system shall persist a user's selected language setting across app sessions.                                                 | 2 | Users should not need to reselect their language every time they open the app.                                              | Nate Sheffield   |  |
| <b>Legal Requirements</b>                       |                                  |                                                                                                                                  |   |                                                                                                                             |                  |  |
| 73                                              | ADA Compliance                   | The system shall comply with the accessibility requirements of the Americans with Disabilities Act.                              | 1 | ADA compliance is a legal obligation for a platform serving the public, including people with disabilities.                 | Tavien Smith     |  |
| 74                                              | Data Breach Notification         | The system shall comply with South Carolina's data breach notification law.                                                      | 1 | State law requires timely notification to users if their personal data is compromised.                                      | Tavien Smith     |  |
| 75                                              | Minor Data Protection            | The system shall obtain parental consent before collecting personal information from a user under the age of 13.                 | 1 | Federal law requires parental consent before collecting data from children under 13.                                        | Tavien Smith     |  |
| 76                                              | Safety Status Data Retention     | The system shall retain a user's safety-status history for 30 days after an event, after which it shall be automatically purged. | 1 | Retaining status history briefly supports post-event reporting, while automatic deletion limits long-term privacy exposure. | Nate Sheffield   |  |